

ISBELL FIELD, AL, USA

WMO: 722111

Lat: 34.474N

Lon: 85.721W

Elev: 267

StdP: 98.15

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 63847

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.6	-5.3	-16.2	0.9	-1.3	-13.0	1.3	-0.7	8.5	5.2	7.7	5.9	1.2	0	0.367

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.5	34.2	22.8	32.9	22.5	32.1	22.3	25.1	30.9	24.5	29.9	23.9	29.1	2.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	18.8	27.2	22.8	18.1	26.7	22.4	17.7	26.3	77.7	30.9	75.1	30.0	72.6	29.1	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.3	5.9	5.2	-11.0	35.8	2.6	2.1	-12.9	37.2	-14.4	38.4	-15.8	39.6	-17.7	41.1		
(5)			-11.6	25.7	2.6	1.0	-13.5	26.5	-15.0	27.1	-16.4	27.7	-18.2	28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.7	4.9	6.9	10.7	15.2	19.7	24.0	25.2	25.4	22.4	16.3	10.1	7.0		
	DBStd	8.41	5.74	5.22	4.96	4.14	3.70	2.15	2.03	2.27	3.02	4.56	4.63	5.27		
	HDD10.0	537	178	114	52	8	0	0	0	0	0	6	54	125		
	HDD18.3	1814	416	322	239	108	31	0	0	0	6	93	248	351		
	CDD10.0	2619	21	26	74	164	300	421	469	479	373	201	58	33		
	CDD18.3	856	0	1	3	15	73	171	212	221	128	31	2	1		
	CDH23.3	7941	0	2	45	215	724	1646	1857	2038	1122	277	13	1		
	CDH26.7	3136	0	0	3	33	204	706	791	914	426	58	1	0		
(14) Wind	WSAvg	1.6	1.9	2.0	2.1	2.0	1.5	1.3	1.2	1.1	1.1	1.4	1.6	1.8		
(15) Precipitation	PrecAvg	1398	137	124	157	118	121	102	111	97	107	82	113	130		
	PrecMax	1673	244	293	268	207	329	223	227	187	220	226	260	209		
	PrecMin	879	52	52	57	67	18	34	47	28	12	2	48	63		
	PrecStd	220	45	54	52	37	66	53	47	38	59	59	54	46		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.1	22.8	27.2	29.7	32.3	35.4	35.7	36.5	34.6	30.9	25.2	21.9		
		MCWB	15.2	16.4	17.0	17.7	21.6	22.1	23.1	23.7	20.4	18.8	16.3	16.4		
	2%	DB	18.0	20.3	24.8	27.6	30.3	33.1	33.2	34.3	32.6	27.8	22.4	19.9		
		MCWB	14.3	15.2	16.1	17.3	20.8	22.2	23.2	23.5	20.7	18.9	15.2	16.4		
	5%	DB	17.0	17.9	22.3	26.1	28.9	32.3	32.3	32.8	31.1	26.4	20.8	17.7		
		MCWB	13.4	12.7	14.6	16.6	20.1	21.8	22.9	23.0	21.2	17.9	14.9	14.6		
	10%	DB	14.8	16.3	20.1	24.0	27.5	31.0	31.1	31.6	29.0	24.8	18.2	16.1		
		MCWB	11.8	11.1	13.6	15.9	18.9	21.8	22.7	22.8	21.1	17.4	13.7	13.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.8	18.0	18.5	20.1	23.3	25.1	25.5	26.1	24.4	22.7	19.3	18.3		
		MCDB	18.5	21.0	23.9	25.9	29.1	30.8	31.1	32.2	29.2	27.8	22.1	20.4		
	2%	WB	15.5	16.7	17.3	19.0	22.3	24.2	24.8	25.3	23.5	21.5	18.0	17.2		
		MCDB	18.1	19.4	22.1	24.4	27.8	30.1	30.3	31.5	28.3	25.3	20.4	18.8		
	5%	WB	14.0	14.7	16.2	18.1	21.5	23.5	24.3	24.6	22.7	20.5	16.6	15.5		
		MCDB	16.5	17.4	20.5	23.4	26.3	29.2	29.4	30.4	27.5	23.9	19.0	17.3		
	10%	WB	12.0	12.5	15.0	17.3	20.5	22.9	23.6	23.9	22.0	19.2	14.8	13.3		
		MCDB	13.8	14.8	18.8	22.3	25.0	28.3	28.6	29.2	26.6	22.9	17.7	15.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	11.6	12.7	14.1	13.0	12.3	11.1	11.5	12.2	13.1	12.8	10.4		
		MCDBR	12.6	14.0	15.6	16.0	14.8	14.7	13.0	13.4	15.0	14.8	15.2	11.7		
	5% WB	MCWBR	9.0	8.9	8.1	7.2	6.1	5.3	4.4	4.4	5.3	6.7	8.7	8.6		
		MCWBR	10.1	11.3	12.5	13.1	11.9	12.0	11.1	11.8	11.0	10.5	10.2	9.7		
(40) Clear-Sky Solar Irradiance	taub		0.316	0.320	0.359	0.394	0.447	0.483	0.535	0.532	0.426	0.374	0.338	0.316		
		taud	2.505	2.489	2.406	2.295	2.189	2.125	1.993	1.990	2.278	2.409	2.455	2.523		
	Ebn at Noon		894	929	914	893	845	811	766	760	833	854	857	872		
		Edh at Noon	84	94	111	131	147	156	178	175	125	101	87	78		
(44) All-Sky Solar Radiation	RadAvg		2.50	3.02	4.15	5.38	5.91	6.21	5.89	5.49	4.73	3.89	2.83	2.12		
	RadStd		0.18	0.39	0.38	0.35	0.46	0.47	0.38	0.29	0.44	0.50	0.32	0.21		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)		+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+144	+105

Nomenclature: See separate page